

Lecture #27

Content to be covered

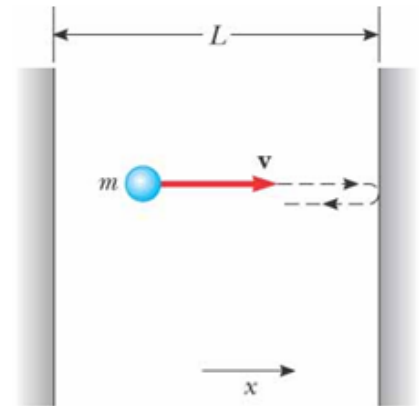
- ❖ Particle in a 1-D Box
 - ❑ Wavefunction
 - ❑ Energy eigen values

Resources to be consulted

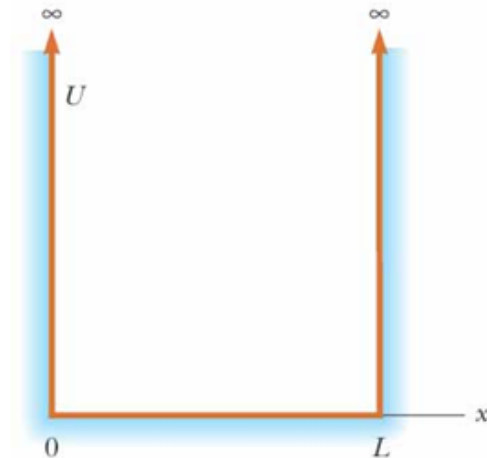
- ❑ Concepts of Modern Physics by Arthur Beiser

Particle in a one dimensional Box (infinite square well potential)

- A particle is confined to a one-dimensional region of space between two **impenetrable** walls separated by distance L . This is a **one- dimensional “box”**
- The particle is bouncing elastically back and forth between the walls.
 - As long as the particle is inside the box, the potential energy does not depend on its location. We can choose this energy value to be zero
- $V= 0, 0 < x < L, V \rightarrow \infty, x \leq 0$ and $x \geq L$



(a)



Particle in a one dimensional Box

- Since the walls are impenetrable, there is zero probability of finding the particle outside the box. Zero probability means that $\psi(x) = 0$, for $x < 0$ and $x > L$
- The wave function must also be 0 at the walls ($x = 0$ and $x = L$), since the wavefunction must be continuous or mathematically, $\psi(0) = 0$ and $\psi(L) = 0$
- In the region $0 < x < L$, where $V = 0$, the Schrödinger equation can be expressed in the form
$$\frac{\hbar^2 \partial^2 \psi(x)}{2m \partial x^2} + V(x) = E \psi(x)$$
- In the region $0 < x < L$, where $V = 0$, the Schrödinger equation can be expressed in the form
$$\frac{\hbar^2 \partial^2 \psi(x)}{2m \partial x^2} = E \psi(x)$$
- We can re-write it as
$$\frac{\partial^2 \psi(x)}{\partial^2 x} = -\frac{2mE}{\hbar^2} \psi(x); \quad \text{where} \quad k^2 \equiv \frac{2mE}{\hbar^2}$$

Particle in a 3-D Box

- Now consider a particle of mass m and total energy E is confined in three dimensional box of x dimension length L_x , y direction L_y and z direction length L_z .
- The particle moves freely within these lengths, as potential is zero inside the box.
- This box is bounded by a large potential barrier.
- The particle is confined in 3 dimensional box, and restricted to outside by a very large potential barriers.

Total Wave-function:
$$\psi(x, y, z) = \sqrt{\frac{2}{L_x}} \sqrt{\frac{2}{L_y}} \sqrt{\frac{2}{L_z}} \sin\left(\frac{n_x \pi}{L_x} x\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right)$$

$$E_x = \frac{n_x^2 \pi^2 \hbar^2}{2mL_x^2}; E_y = \frac{n_y^2 \pi^2 \hbar^2}{2mL_y^2}; E_z = \frac{n_z^2 \pi^2 \hbar^2}{2mL_z^2}$$

Total Energy:
$$E = \frac{\pi^2 \hbar^2}{2m} \left[\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} + \frac{n_z^2}{L_z^2} \right]$$

Particle in a one dimensional Box

The most general solution to this differential equation is given by

$\psi(x) = A \sin(kx) + B \cos(kx)$ where A and B are constants determined by the properties of the wavefunction as well as boundary and normalization conditions.

- Since the particle can not have infinite energy, it cannot exist outside the box. Therefore the wavefunction $\psi(x)$ must be zero outside the box. $\psi(x)$ must be also zero at the walls, that is at $x = 0$ & $x = L$, for otherwise there would be discontinuities at the walls.
- $\sin(kx)$ and $\cos(kx)$ are finite and single-valued functions.
- Continuity : $\psi(x=0) = \psi(x=L) = 0$

which means $\psi(x=0) = A \sin k0 + B \cos k0 \Rightarrow B = 0 \Rightarrow \psi(x) = A \sin(kx)$

$$\psi(x=L) = A \sin kL \Rightarrow kL = \pm n\pi \quad n = \pm 1, \pm 2, \pm 3, \dots$$

Here, **$n = 0$ is not admissible** because it yields ψ zero everywhere which means that the particle is nowhere. So, $\psi(x) = A \sin\left(\frac{n\pi}{L}x\right)$

Normalizing Ψ : $\int_0^L \psi^* \psi dx = 1 \Rightarrow A = \sqrt{\frac{2}{L}}$

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right)$$

Particle in a one dimensional Box

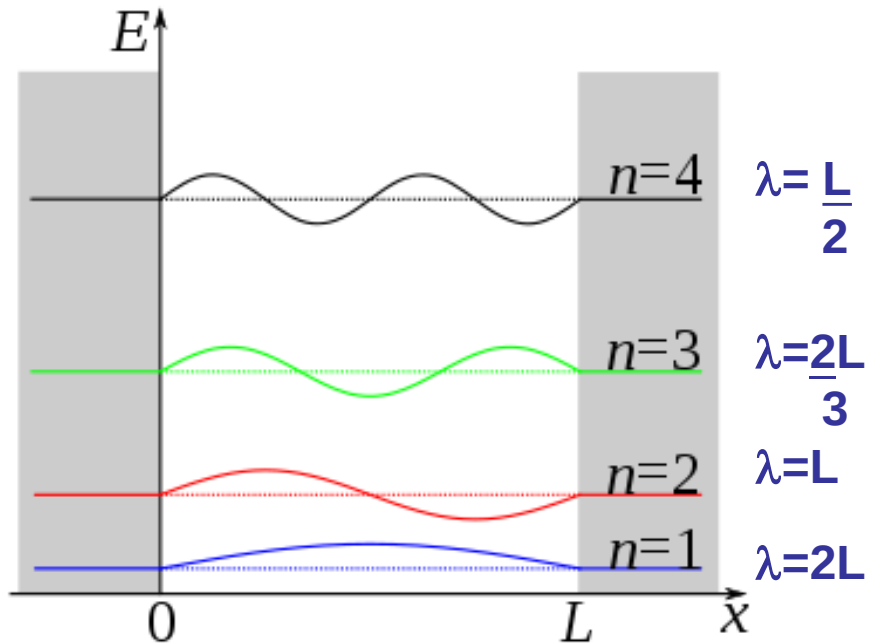
$$k_n^2 = \frac{2mE_n}{\hbar^2} \Rightarrow E_n = \frac{\hbar^2 k_n^2}{2m} \quad \text{but } k_n = \frac{\pi}{L} n$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} = \frac{n^2 h^2}{8mL^2}$$

$$E_1 = \frac{h^2}{8mL^2}; E_2 = \frac{2h^2}{8mL^2}; E_3 = \frac{9h^2}{8mL^2}; \dots$$

- A trapped particle cannot have an arbitrary energy, as a free particle can, *but can have certain discrete energy corresponding to $n=1,2,3,\dots$*
- Each permitted energy is called 'eigen value' of the particle and constitute the energy level of the system, and the integer ' n ' that specifies an energy level E_n , is called its quantum number. The wave function $\psi(x)$ corresponding to each eigen value is called eigen functions.
- A trapped particle cannot have zero energy.
- The lowest energy for a quantum system is termed as *ground state*. $E_1 = h^2/8mL^2 > 0$, whereas the corresponding classical system would have a minimum energy of zero.

Particle in a one dimensional Box



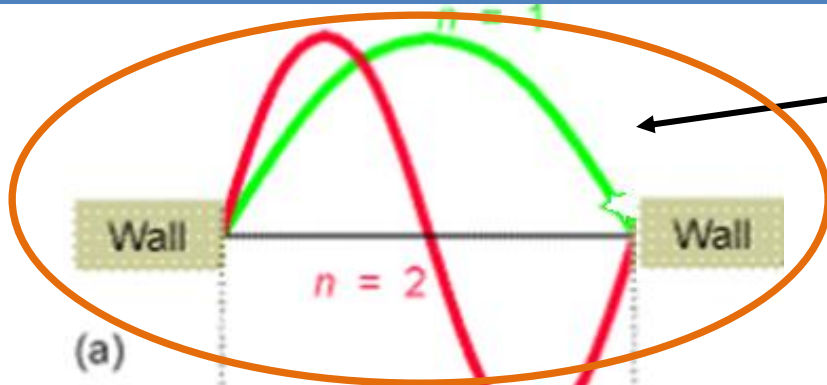
$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L}\right)x,$$

$$\psi_2(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi}{L}\right)x,$$

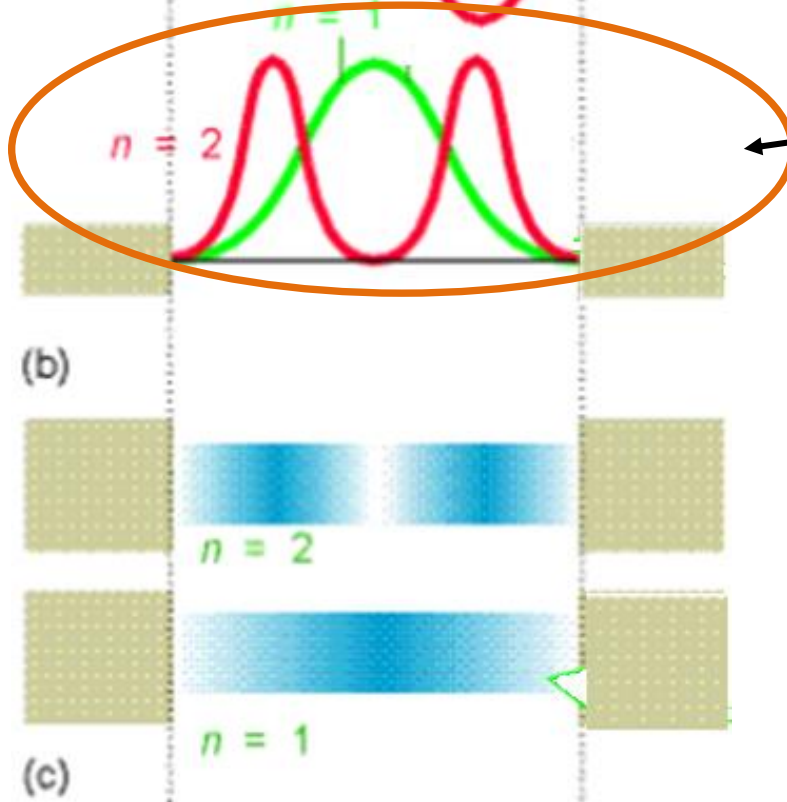
$$\psi_3(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{3\pi}{L}\right)x$$

$\vdots$

Particle in a one dimensional Box

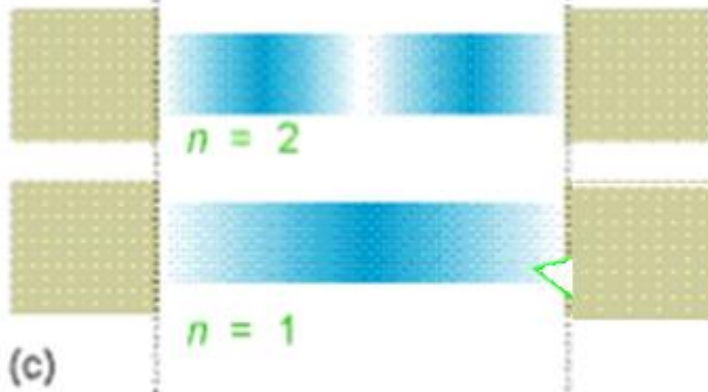


$$\psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$



Probability distribution : $\psi_n^* \psi_n$
with $n=1,2,3,\dots$

Note: Particle most likely to be found in the middle for $n=1$.



(a) The first two wavefunctions, (b) the corresponding probability distributions, and (c) a representation of the probability distribution in terms of the darkness of shading.

Problems

Consider an electron is in the 10nm wide well with infinite barriers. Assume that a photon is absorbed, and the electron is transferred from the ground state to the second excited state. What was the wavelengths of the photon?

ground state ($n = 1$) and second excited state ($n = 3$)

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

$$E_3 = \frac{3^2 \pi^2 \hbar^2}{2mL^2}$$

$$E_{\text{ground}} = E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$$

Third excited state is $E_3 : E_3 = E_1 \times 3^2 = 9 \times 0.00375 \text{ eV} \approx 0.0338 \text{ eV}$

$$E_3 - E_1 = 0.0338 - 0.00375 \approx 0.030 \text{ eV}$$

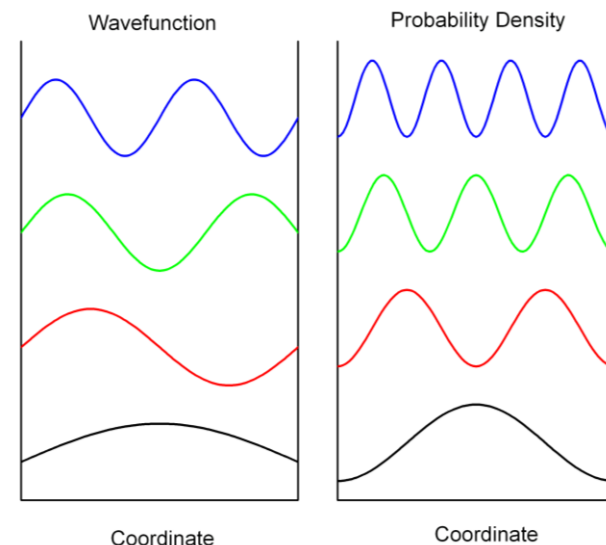
$$\lambda = \frac{1240}{0.03} \approx 41 \mu\text{m}$$

Problems

1. Probability to find particle in the right half of the well

$$\int_{L/2}^L |\Psi(x)|^2 dx = \int_{L/2}^L \left[\sqrt{\frac{2}{L}} \sin kx \right]^2 dx \Rightarrow \frac{2}{L} \int_{L/2}^L \sin^2(kx) dx = \frac{2}{L} \left[\frac{L}{2} \frac{1}{2} \right] = \frac{1}{2}$$

2. An electron is confined to an one-dimensional infinity potential well of width 0.2 nm. It is found that when the energy of the particle is 230 eV, its eigenfunction has 5 antinodes. Find the mass of the particle and show that it can never have energy equal to 1 keV. $m = 9.1 \times 10^{-31}$ Kg, $n = 10.4$



3. A 10-g marble is in a box 10 cm across. Find its permitted energies.
4. Find the probability that a particle trapped in a box L wide can be found between $0.45L$ and $0.55L$ for the ground and first excited states.
5. Find the expectation value x of the position of a particle trapped in a box L wide.
6. Calculate the energy difference between the ground state and the first excited state for an electron in one dimensional rigid box of length $10 \text{ \AA}$.